DBMS – DATABASE RELATIONSHIP ANALYSIS USING JOINS

Project Title: Alibaba B2B E-Commerce Database Management System

Skill Task: DBMS

Week: 8

1. Combining Multiple Tables

1.1 Customer, Orders, Order_Item and Product

Combines four related tables to show which products each customer ordered.

```
SQL> SELECT c.Customer_ID, c.Business_Name, o.Order_ID, o.Order_Date,
  2         p.Product_Name, od.Quantity, od.Unit_Price
  3  FROM Customer c
  4  INNER JOIN Orders o ON c.Customer_ID = o.Customer_ID
  5  INNER JOIN Order_Details od ON o.Order_ID = od.Order_ID
  6  LEFT JOIN Product p ON od.Product_ID = p.Product_ID
  7  ORDER BY o.Order_Date, o.Order_ID;

CUSTOMER_ID BUSINESS_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                        QUANTITY
--------- -------------------------------------------------- ----------
UNIT_PRICE
----------
       101 ABC Traders                                              1001
01-SEP-26                                                            8
      5000

       101 ABC Traders                                              1001
01-SEP-26                                                            5
      5000

CUSTOMER_ID BUSINESS_NAME                                      ORDER_ID
```

```
CUSTOMER_ID BUSINESS_NAME                                          ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                           QUANTITY
--------- -------------------------------------------------- ----------
UNIT_PRICE
----------

        102 Tech Solutions                                           1002
02-SEP-26                                                            10
      1850

        102 Tech Solutions                                           1002
02-SEP-26                                                            10

CUSTOMER_ID BUSINESS_NAME                                          ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                           QUANTITY
--------- -------------------------------------------------- ----------
UNIT_PRICE
----------
      1850

        103 Green Mart                                               1003
```

2. INNER JOIN

2.1 Customers and Orders

Displays only customers who have placed orders.

```
SQL> SELECT c.Customer_ID, c.Business_Name, o.Order_ID, o.Order_Date
  2  FROM Customer c
  3  INNER JOIN Orders o ON c.Customer_ID = o.Customer_ID
  4  ORDER BY o.Order_Date;

CUSTOMER_ID BUSINESS_NAME                                    ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        101 ABC Traders                                            1001
01-SEP-26

        102 Tech Solutions                                         1002
02-SEP-26

        103 Green Mart                                             1003
03-SEP-26
```

```
CUSTOMER_ID BUSINESS_NAME                                          ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        104 Royal Exports                                            1004
04-SEP-26

        105 Sun Enterprises                                          1005
05-SEP-26

        101 ABC Traders                                              1006
06-SEP-26
```

2.2 Orders and Order_details

Displays each order with its corresponding Order_details.

```
SQL> SELECT o.Order_ID, o.Order_Date, od.Product_ID, od.Quantity, od.Unit_Price,
  2         (od.Quantity * od.Unit_Price) AS Line_Total
  3  FROM Orders o
  4  INNER JOIN Order_Details od ON o.Order_ID = od.Order_ID
  5  ORDER BY o.Order_ID;

  ORDER_ID ORDER_DAT PRODUCT_ID   QUANTITY UNIT_PRICE LINE_TOTAL
---------- --------- ---------- ---------- ---------- ----------
      1001 01-SEP-26        301          8       5000      40000
      1001 01-SEP-26        301          5       5000      25000
      1002 02-SEP-26        302         10       1850      18500
      1002 02-SEP-26        302         10       1850      18500
      1003 03-SEP-26        303          8       4000      32000
      1003 03-SEP-26        303          8       4000      32000
      1004 04-SEP-26        304         15        850      12750
      1004 04-SEP-26        304          8        900       7200
      1005 05-SEP-26        305         10       4500      45000
      1005 05-SEP-26        305         10       4500      45000
      1006 06-SEP-26        306          7       1400       9800

  ORDER_ID ORDER_DAT PRODUCT_ID   QUANTITY UNIT_PRICE LINE_TOTAL
---------- --------- ---------- ---------- ---------- ----------
      1006 06-SEP-26        306          7       1400       9800
      1007 07-SEP-26        307          5       4300      21500
      1007 07-SEP-26        307          7       4300      30100
      1008 08-SEP-26        308         10       3650      36500
      1008 08-SEP-26        308         10       3650      36500
      1009 09-SEP-26        309          7       2200      15400
      1009 09-SEP-26        309          7       2200      15400
      1010 10-SEP-26        310         17       1700      28900
      1010 10-SEP-26        310         17       1700      28900

20 rows selected.
```

3. LEFT JOIN

3.1 Display All Customers and Their Orders

Displays all customers, including customers who have not placed an order.

```
SQL> SELECT c.Customer_ID, c.Business_Name, o.Order_ID, o.Order_Date
  2  FROM Customer c
  3  LEFT JOIN Orders o ON c.Customer_ID = o.Customer_ID
  4  ORDER BY c.Customer_ID, o.Order_Date;

CUSTOMER_ID BUSINESS_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        101 ABC Traders                                              1001
01-SEP-26

        101 ABC Traders                                              1006
06-SEP-26

        102 Tech Solutions                                           1002
02-SEP-26


CUSTOMER_ID BUSINESS_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        102 Tech Solutions                                           1007
07-SEP-26

        103 Green Mart                                               1003
03-SEP-26

        103 Green Mart                                               1008
08-SEP-26
```

4. RIGHT JOIN

4.1 Display All Orders and Customer Details

Displays all orders along with the matching customer information.

```
SQL> SELECT c.Customer_ID, c.Business_Name, o.Order_ID, o.Order_Date
  2  FROM Customer c
  3  RIGHT JOIN Orders o ON c.Customer_ID = o.Customer_ID
  4  ORDER BY o.Order_Date;

CUSTOMER_ID BUSINESS_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        101 ABC Traders                                              1001
01-SEP-26

        102 Tech Solutions                                           1002
02-SEP-26

        103 Green Mart                                               1003
03-SEP-26


CUSTOMER_ID BUSINESS_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        104 Royal Exports                                            1004
04-SEP-26

        105 Sun Enterprises                                          1005
05-SEP-26

        101 ABC Traders                                              1006
06-SEP-26
```

5. Complete Order Details

Combines Customer, Orders, Order_Item, Product, Supplier and Payment tables.

```
SQL> SELECT c.Customer_ID, c.Business_Name, o.Order_ID, o.Order_Date,
  2         od.Product_ID, p.Product_Name, od.Quantity, od.Unit_Price,
  3         (od.Quantity * od.Unit_Price) AS Line_Total
  4  FROM Customer c
  5  INNER JOIN Orders o ON c.Customer_ID = o.Customer_ID
  6  INNER JOIN Order_Details od ON o.Order_ID = od.Order_ID
  7  LEFT JOIN Product p ON od.Product_ID = p.Product_ID
  8  ORDER BY o.Order_Date DESC, o.Order_ID;

CUSTOMER_ID BUSINESS_NAME                                          ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID PRODUCT_NAME
--------- ---------- --------------------------------------------------
  QUANTITY UNIT_PRICE LINE_TOTAL
---------- ---------- ----------
       105 Sun Enterprises                                          1010
10-SEP-26        310
       17       1700      28900

       105 Sun Enterprises                                          1010
10-SEP-26        310
       17       1700      28900

CUSTOMER_ID BUSINESS_NAME                                          ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID PRODUCT_NAME
--------- ---------- --------------------------------------------------
  QUANTITY UNIT_PRICE LINE_TOTAL
---------- ---------- ----------

       104 Royal Exports                                            1009
09-SEP-26        309
         7       2200      15400

       104 Royal Exports                                            1009
```

6. Customer Purchase History

Displays the products purchased by each customer.

```
SQL> SELECT c.Customer_ID, c.Business_Name, o.Order_ID, o.Order_Date,
  2         od.Product_ID, p.Product_Name, od.Quantity,
  3         (od.Quantity * od.Unit_Price) AS Line_Total
  4  FROM Customer c
  5  INNER JOIN Orders o ON c.Customer_ID = o.Customer_ID
  6  INNER JOIN Order_Details od ON o.Order_ID = od.Order_ID
  7  LEFT JOIN Product p ON od.Product_ID = p.Product_ID
  8  ORDER BY c.Customer_ID, o.Order_Date;

CUSTOMER_ID BUSINESS_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID PRODUCT_NAME
--------- ---------- --------------------------------------------------
  QUANTITY LINE_TOTAL
---------- ----------
       101 ABC Traders                                             1001
01-SEP-26        301
         5      25000

       101 ABC Traders                                             1001
01-SEP-26        301
         8      40000

CUSTOMER_ID BUSINESS_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID PRODUCT_NAME
--------- ---------- --------------------------------------------------
  QUANTITY LINE_TOTAL
---------- ----------

       101 ABC Traders                                             1006
06-SEP-26        306
         7       9800
```

7. Customer Purchase History with Payment

Displays each customer purchase together with its payment method and status.

```
SQL> SELECT c.Customer_ID,
  2         c.Business_Name,
  3         o.Order_ID,
  4         o.Order_Date,
  5         p.Product_Name,
  6         od.Quantity,
  7         (od.Quantity * od.Unit_Price) AS Amount_Spent,
  8         py.Payment_Method,
  9         py.Payment_Status
 10  FROM   Customer c
 11         INNER JOIN Orders o        ON c.Customer_ID = o.Customer_ID
 12         INNER JOIN Order_Details od ON o.Order_ID = od.Order_ID
 13         LEFT JOIN Product p        ON od.Product_ID = p.Product_ID
 14         LEFT JOIN Payment py       ON o.Order_ID = py.Order_ID
 15  ORDER BY c.Customer_ID, o.Order_Date;

CUSTOMER_ID BUSINESS_NAME                                       ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                        QUANTITY
--------- -------------------------------------------------- ----------
AMOUNT_SPENT PAYMENT_METHOD
------------ --------------------------------------------------
PAYMENT_STATUS
--------------------
         101 ABC Traders                                            1001
01-SEP-26                                                           5
        25000 credit card
successful
```

```
CUSTOMER_ID BUSINESS_NAME                                       ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                        QUANTITY
--------- -------------------------------------------------- ----------
AMOUNT_SPENT PAYMENT_METHOD
------------ --------------------------------------------------
PAYMENT_STATUS
--------------------
        101 ABC Traders                                            1001
01-SEP-26                                                           8
       40000 credit card
successful


CUSTOMER_ID BUSINESS_NAME                                       ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                        QUANTITY
--------- -------------------------------------------------- ----------
AMOUNT_SPENT PAYMENT_METHOD
------------ --------------------------------------------------
PAYMENT_STATUS
--------------------
        101 ABC Traders                                            1006
06-SEP-26                                                           7
        9800 credit card
pending
```

8. Multi-Table Business Report

```
SQL> SELECT c.Customer_ID, c.Business_Name,
  2         COUNT(DISTINCT o.Order_ID) AS Total_Orders,
  3         SUM(od.Quantity) AS Total_Units,
  4         SUM(od.Quantity * od.Unit_Price) AS Total_Purchase
  5  FROM Customer c
  6  INNER JOIN Orders o ON c.Customer_ID = o.Customer_ID
  7  INNER JOIN Order_Details od ON o.Order_ID = od.Order_ID
  8  GROUP BY c.Customer_ID, c.Business_Name
  9  ORDER BY Total_Purchase DESC;

CUSTOMER_ID BUSINESS_NAME                                      TOTAL_ORDERS
----------- -------------------------------------------------- ------------
TOTAL_UNITS TOTAL_PURCHASE
----------- --------------
        105 Sun Enterprises                                               2
         54         147800

        103 Green Mart                                                    2
         36         137000

        102 Tech Solutions                                                2
         32          88600


CUSTOMER_ID BUSINESS_NAME                                      TOTAL_ORDERS
----------- -------------------------------------------------- ------------
TOTAL_UNITS TOTAL_PURCHASE
----------- --------------
        101 ABC Traders                                                   2
         27          84600

        104 Royal Exports                                                 2
         37          50750
```

9. Payment Transaction Report

Displays every order with its payment transaction details, latest first.

```
SQL> SELECT c.Business_Name,
  2         o.Order_ID,
  3         o.Order_Date,
  4         (SELECT SUM(od.Quantity * od.Unit_Price)
  5            FROM Order_Details od
  6           WHERE od.Order_ID = o.Order_ID) AS Order_Total,
  7         py.Payment_Method,
  8         py.Amount,
  9         py.Payment_Status
 10  FROM   Customer c
 11         INNER JOIN Orders o    ON c.Customer_ID = o.Customer_ID
 12         LEFT JOIN Payment py   ON o.Order_ID = py.Order_ID
 13  ORDER BY o.Order_Date DESC;

BUSINESS_NAME                                     ORDER_ID ORDER_DAT
------------------------------------------------- ---------- ---------
ORDER_TOTAL PAYMENT_METHOD                                      AMOUNT
----------- ------------------------------------------------- ----------
PAYMENT_STATUS
--------------------
Sun Enterprises                                         1010 10-SEP-26
      57800 credit card                                           60.5
refunded

Royal Exports                                           1009 09-SEP-26
      30800 bank transfer                                          300
successful
```

```
BUSINESS_NAME                                      ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
ORDER_TOTAL PAYMENT_METHOD                                        AMOUNT
----------- -------------------------------------------------- ----------
PAYMENT_STATUS
--------------------

Green Mart                                            1008 08-SEP-26
      73000 debit card                                            45.25
failed

Tech Solutions                                        1007 07-SEP-26
      51600 paypal                                                 150

BUSINESS_NAME                                      ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
ORDER_TOTAL PAYMENT_METHOD                                        AMOUNT
----------- -------------------------------------------------- ----------
PAYMENT_STATUS
--------------------
successful

ABC Traders                                           1006 06-SEP-26
      19600 credit card                                           89.99
pending

Sun Enterprises                                       1005 05-SEP-26
```